***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 11   Genetics of Bacteria and Archaea

11.1   Multiple Choice Questions

1) A mutant that has a nutritional requirement for growth is an example of a(n)
A) autotroph.
B) auxotroph.
C) heterotroph.
D) organotroph.
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.1

2) Consider a mutation in which the change is from UAC to UAU. Both codons specify the amino acid tyrosine. Which type of point mutation is this?
A) silent mutation
B) nonsense mutation
C) missense mutation
D) frameshift mutation
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.2

3) A mutation that readily reverses to restore the original parental type would most likely be due to a(n)
A) deletion.
B) insertion.
C) point mutation.
D) frameshift mutation.
Answer: C
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.2

4) Which process listed below allows genetic material to be transferred from a virus-like particle that lacks genes for its own replication?
A) conjugation of an F+ plasmid
B) gene transfer through a gene transfer agent
C) transduction by a dsDNA phage Mu
D) transformation of a linear piece of DNA
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.5, 11.7

5) The mutagens 2-aminopurine and 5-bromouracil are examples of
A) alkylating agents.
B) nucleotide base analogs.
C) chemicals reacting with DNA.
D) None of the answers are correct.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

6) The killing of cells by UV irradiation involves
A) absorption at 260 nm by proteins only.
B) absorption at 260 nm by RNA only.
C) formation of pyrimidine dimers.
D) formation of purine dimers.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

7) Ionizing radiation does NOT include
A) gamma rays.
B) UV rays.
C) X-rays.
D) cosmic rays.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

8) Which of the following methods may introduce foreign DNA into a recipient?
A) transformation
B) transduction
C) conjugation
D) transformation, transduction, and conjugation
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.8

9) The uptake of free DNA from the environment ________, while transfer of DNA with cell-to-cell contact would most likely result in ________.
A) transformation / conjugation
B) transduction / conjugation
C) conjugation / transformation
D) transformation / transduction
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.8

10) Which of the following is NOT required for homologous recombination?
A) an Hfr chromosome
B) RecA
C) proteins having helicase activity
D) endonuclease
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.5

11) Consider the following experiment. First, large populations of two mutant strains of *Escherichia coli* are mixed, each requiring a different, single amino acid. After plating them onto a minimal medium, 45 colonies grew. Which of the following may explain this result?
A) The colonies may be due to back mutation (reversion).
B) The colonies may be due to recombination.
C) Either A or B is possible.
D) Neither A nor B is possible.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.3, 11.5

12) You have performed the following mating experiment using Hfr and F-strains of *Escherichia coli*:
Hfr (thr^{+} leu^{+} gal^{+} str^{s}) × F^{-} (thr^{-} leu^{-} gal^{-} str^{r}). Which of the following selective media would you use to score recombinant colonies?
A) minimal medium
B) minimal medium + streptomycin
C) minimal medium + threonine
D) minimal medium + streptomycin + threonine
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.9

13) Horizontal gene transfer in *Archaea*
A) is infrequent in nature and therefore difficult to use for genetic studies in the laboratory.
B) has not been documented, thus all genetic studies of archaea are done via genomic sequencing.
C) frequently occurs in nature and has been used to perform genetic studies in the laboratory as well.
D) frequently occurs in nature, but there are very few laboratory studies because archaea do not cause human disease.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.10

14) The minimal amount of genetic information required for specialized transduction would include
A) the *att* region.
B) the *cos* site.
C) a helper phage.
D) the *att* region, *cos* site, and a helper phage.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.7

15) Lysogeny probably carries a strong selective advantage for the host cell because it
A) prevents cell lysis.
B) confers resistance to infection by viruses of the same type.
C) confers resistance to infection by viruses of a different type (or strain).
D) confers resistance to infection by many virus types and prevent cell lysis.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.7

16) A plasmid may
A) replicate independently of the chromosome.
B) be transferred cell-to-cell during conjugation.
C) be integrated into the chromosome.
D) replicate independently of the chromosome, integrate into the chromosome, or be transferred cell-to-cell during conjugation.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.9

17) Plasmids that govern their own transfer are known as
A) transformable.
B) transmutable.
C) conjugative.
D) transfective.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.8

18) Homologous recombination has been observed in
A) *Archaea*.
B) *Bacteria*.
C) *Eukarya*.
D) All three domains *(Archaea*, *Bacteria*, and *Eukarya)*.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.5

19) Hfr strains of *Escherichia coli*
A) do not possess an F factor.
B) have the F factor as a plasmid.
C) have an integrated F factor.
D) transfer the complete F factor to recipient cells at a high frequency.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.9

20) F^+ strains of *Escherichia coli*
A) do not have an F factor.
B) have the F factor as a plasmid.
C) have an integrated F factor.
D) transfer the F factor to recipient cells at a high frequency.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.8

21) F^-strains of *Escherichia coli*
A) do not have an F factor.
B) have the F factor as a plasmid.
C) have an integrated F factor.
D) transfer the F factor to other strains at a high frequency.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.8

22) All Hfr strains integrate into the chromosome at
A) the same locus.
B) several specific sites.
C) the same locus most of the time, although there may be some variation.
D) loci that cannot be accurately determined.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.9

23) Transposition is a(n)
A) homologous recombination event.
B) analogous recombination event.
C) site-specific recombination event.
D) general recombination event.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.11

24) The enzyme transposase may be coded for by insertion sequences on a
A) chromosome.
B) phage.
C) plasmid.
D) chromosome, phage, or plasmid.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.11

25) A ″point mutation″ refers to mutations involving
A) a base-pair substitution.
B) the gain of a base pair (microinsertion).
C) the deletion of a base pair (microdeletion).
D) a substitution, deletion, or addition of one base-pair.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.2

26) A deleterious mutation in *recA* results in
A) a decrease in specific recombination.
B) a decrease in homologous recombination.
C) an increase in homologous recombination.
D) no change in either general or specific recombination.
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.5

27) The production of a functional gene product by transforming bacteria that lack a *lacZ* gene with a plasmid containing a *lacZ* gene is known as
A) complementation.
B) mitosis.
C) transfection.
D) reversion.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.5

28) Consider conjugation in *Escherichia coli*. In which of the following matings would chromosomal genes be transferred most frequently?
A) $F^+ \times F^-$
B) $F^- \times F^-$
C) Hfr × F^-
D) F^+ x F^+
Answer: C
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.9

29) Which of the following features are common to transformation, transduction, and conjugation?
(1) unidirectional transfer of genes
(2) incomplete gene transfer
(3) homologous recombination
(4) meiosis occurring in the recipient
A) 1, 2, 3
B) 1, 2
C) 3, 4
D) 1, 2, 4
Answer: A
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.5

30) Which of the following is most similar to lysogeny?
A) Hfr state
B) F^+ state
C) F^- state
D) F' state
Answer: A
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.9

31) In the bacterial world, a gene located on which of the following would be the LEAST likely to be transferred?
A) a resistance plasmid
B) An F plasmid
C) the phage Mu
D) the chromosome
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.9

32) Genetic recombination involving insertion sequences typically results in what type of mutation?
A) base-pair substitution mutation
B) silent mutation
C) frameshift mutation
D) base-pair deletion mutation
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.11

33) The SOS regulatory system is activated by
A) the activity of DNA polymerase IV.
B) DNA damage.
C) transcription of LexA.
D) repression of RecA.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

34) Which of the following factors has delayed the development of laboratory-based genetic systems in *Archaea*?
A) There are no documented systems of conjugation in *Archaea*.
B) Homologous recombination does NOT occur in *Archaea*.
C) Archaea do NOT host viruses or plasmids.
D) Many archaea grow in extreme or unusual conditions that make the use of agar and traditional mutant screening techniques problematic.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.10

35) Transformation and homologous recombination allow for the formation of heteroduplex DNA. Which of the following would occur during DNA replication of this molecule?
A) One daughter strand is complementary to the recombinant DNA molecule, while the other daughter strand is complementary to the parent DNA molecule.
B) Both daughter strands are complementary to the recombinant DNA molecule.
C) Both daughter strands are complementary to the parent DNA molecule.
D) None of the answers are correct.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.6

36) You work for a biotechnology company that uses *Streptomyces* strains to produce pharmaceutical products. A phage has infected and killed some of your *Streptomyce*s strains during production, resulting in dramatically decreased yields. To protect the strains from infection you propose to
A) introduce gene transfer agents into the *Streptomyces* cultures to transfer antibiotic resistance genes into your *Streptomyces* strains.
B) design and insert CRISPR spacer sequences into the genomes of your strains that are complementary to the genomes of the phages that are infecting the cultures.
C) infect the *Streptomyces* strains with a helper phage that will help the strains resist infection.
D) transform the *Streptomyces* strains with plasmids encoding antibody proteins that will protect them from phage infection.
Answer: B
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.12

37) Hfr means high frequency of ________, and these cells are capable of transferring genes from their ________ to other cells.
A) transformation / chromosome
B) transduction / plasmids
C) recombination / chromosome
D) transduction / chromosome
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.9

38) Mutation rates are similar in *Bacteria* and *Archaea*, yet certain stressful conditions mutation rates increase. Why is the mutation rate not constant and close to zero all of the time?
A) Increased mutation rates can be advantageous in rapidly changing environments because some random mutations may be useful for survival.
B) Microorganisms carefully control the mutation rate of their DNA to match the environmental conditions and maximize evolution.
C) The increased mutation rate under stressful conditions is an indication that the microorganisms can no longer replicate their DNA properly and are about to die.
D) Constant mutation rates would halt evolution completely.
Answer: A
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.3

39) Chemical mutagens, UV radiation, and ionizing radiation all increase mutation rates, but they have different mechanisms. Which type of mutagen would be best suited for creating large deletions and rearrangements within a genome?
A) chemical mutagens
B) UV radiation
C) ionizing radiation
D) Chemical, UV, and ionizing radiation would create large deletions and rearrangements if used in a very high dose.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.4

40) The SOS system repairs DNA that has gaps, breaks, and other lesions by
A) cutting DNA from other parts of the genome and pasting it into the gaps or damaged areas.
B) stabilizing single-stranded DNA until the next round of normal replication.
C) using specialized DNA polymerases that will synthesize a new DNA strand even if there is not a normal complementary DNA strand to act as a template.
D) using available mRNA and a special RNA-dependent DNA polymerase to fill in the gaps and replace damaged DNA.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

41) Microinsertions and microdeletions often result in ________ mutations.
A) auxotrophic
B) advantageous
C) silent
D) frameshift
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.2

42) When damaged or single-stranded DNA activates the RecA protein, the RecA protein stimulates the cleavage of LexA. This results in
A) repression of polymerase V and activation of endonuclease.
B) activation of the Hfr system.
C) derepression of the SOS system.
D) increased transduction and recombination.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

43) The F (fertility) plasmid contains a set of genes that encode for the ________ proteins that are essential in conjugative transfer of DNA.
A) pili
B) SOS repair
C) transduction
D) transformation
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.8

44) The designations Phe-, Leu-, and Ser+ refer to an organism′s
A) plasmid type.
B) genotype.
C) phenotype.
D) mutation type.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.1

45) If a bacterium carrying a plasmid that confers resistance to ampicillin is placed into medium without ampicillin, it may
A) gain resistance to other antibiotics.
B) transfer resistance to other cultures in the laboratory.
C) undergo a reversion mutation.
D) lose the plasmid because there is no selection for ampicillin resistance.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.3

46) When DNA is transferred into a prokaryotic cell it may
A) be degraded by enzymes.
B) replicate independent of the host chromosome.
C) recombine with the host chromosome.
D) be degraded by enzymes, replicate independent of the host chromosome, or recombine with the host chromosome.
Answer: D
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.12

47) The process in which related DNA sequences from two different sources are exchanged is called
A) transduction.
B) phage conversion.
C) reversion.
D) homologous recombination.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.5

48) Integration of linear transforming DNA into the chromosome
A) is not required for the expression for the transformed genes.
B) is catalyzed by the RecA gene.
C) almost never occurs because restriction endonuclease will degrade the DNA before it is integrated.
D) only occurs in laboratory-based systems in artificial competent cells.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.6

49) High-efficiency, natural transformation
A) is common in *Bacteria* and *Archaea.*
B) requires specialized DNA uptake, DNA binding, and integration proteins.
C) is only common in *Archaea*.
D) usually involves plasmids.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.6

50) The CRISPR system
A) facilitates homologous recombination through a complex system of proteins and clustered repeats.
B) recognizes foreign DNA sequences that have previously entered the cell and directs the Cas proteins to destroy them.
C) repairs DNA and increases DNA damage tolerance during times of stress.
D) synthesizes gene transfer agents during stationary phase.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.12

51) Which of the following is the correct abbreviation for a mutation in a gene that synthesizes one of the enzymes involved in tryptophan production?
A) *TrpC*
B) trp
C) Trp^-
D) *trpC1*
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.1

52) Which of the following is TRUE of competence?
A) It is not required for transformation.
B) It commonly occurs with high efficiency in nature.
C) It cannot occur naturally in bacteria.
D) It requires special proteins such as a cell wall autolysin.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.6

53) What is one of the changes that occur when a cell contains an F plasmid that is NOT integrated into the chromosome?
A) It is no longer able to produce a pilus.
B) Cell surface receptors change, preventing the uptake of more plasmids through conjugation.
C) Mutation rates are decreased.
D) The cell is considered an Hfr cell.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.9

11.2 True/False Questions

1) UV radiation is a useful tool in producing mutants of microbial cultures.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

2) When UV radiation damage occurs, DNA repair occurs only in the absence of template instruction.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

3) Following uptake, transforming DNA becomes attached to a competence-specific protein that prevents it from nuclease attack until it reaches the chromosome.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.6

4) The evolutionary significance of phage conversion likely stems from an effective alteration of host cells.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.7

5) Bacterial mating (or conjugation) is a bidirectional process where nucleic acids (DNA or RNA) are transferred between two cells.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.8

6) Laboratory-based genetic systems have been difficult to develop for *Archaea*, because they do NOT naturally undergo conjugation or transduction.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.10

7) Almost all plasmids are double-stranded DNA.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.8

8) Most plasmids are circular rather than linear.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.8

9) When the F factor is integrated, all Hfr strains have the origin of replication functioning in the same direction.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.9

10) In a prokaryotic genome, either insertion elements OR transposons are present, but both are never present at the same time.
Answer: FALSE
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.11

11) The use of transposons to generate mutations is a convenient way to create bacterial mutations in the laboratory.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.11

12) Transposons can be found on many genetic elements, including plasmids, chromosomes, and viral genomes.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.11

13) In transformation experiments using a variety of *Bacteria*, it has been noted that essentially all of the cells in a population can become competent.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.6

14) In specialized transduction, virtually any genetic marker can be transferred from donor to recipient.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.7

15) In specialized transduction, as exemplified by lambda phage in *E. coli*, transduction occurs at high efficiency for only a restricted group of genes near the insertion site of lambda.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.7

16) Many *Bacteria* isolated from nature are natural lysogens.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.7

17) Lysogeny is essential for the virulence of many pathogenic bacterial strains.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.7

18) Proteins and nucleic acids absorb light maximally at 260 nm; hence, proteins protect cells from UV effects.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

19) Insertion sequences are found on both ends of transposons and encode for transposase.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.11

20) Toxigenicity in *Corynebacterium diphtheriae* is due to phage conversion.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.7

21) All Hfr strains possess an F factor integrated into the host chromosome.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.9

22) Intercalating agents, like acridine orange and ethidium bromide, lead to mutagenesis by pushing DNA base pairs apart, which can lead to insertions or deletions.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.4

23) A typical mutation rate for a bacterium is in the range of 10^{-6} to 10^{-7} per kbp.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.3

24) *Archaea* have a unique form of conjugation involving cytoplasmic bridges for bidirectional transfer of DNA.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.11

25) A phage can be infectious even if all of its DNA has been replaced by bacterial DNA.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.7

11.3 Essay Questions

1) Design an experiment (including controls) to determine if a new kitchen chemical cleaner is mutagenic. Include in your answer how you would interpret the results.
Answer: The Ames test could be used to determine if the cleaner solution causes mutations in bacteria. An auxotrophic mutant strain (e.g., *Salmonella enterica* His^-) containing a point mutation is most helpful for this test, so it can measurably revert in small populations of cells. A His^-cell population exposed to various concentrations of the cleaner solution would then be put into agar plates lacking histidine, and the number of colonies that grew would represent cells that mutated back to wildtype (His^+). An important control for the test is to plate an equal number of cells not exposed to the cleaner solution on to the same medium to calculate a baseline mutation rate of the cells when not exposed to the chemical cleaner. If there are significantly more His^+ revertants that grew after exposure to the cleaner solution, then the chemical is mutagenic.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.2

2) Explain how electroporation works and the significance of the procedure for genetic experiments.
Answer: Electroporation involves pulsing cells with a high voltage. The brief electrical charge makes cells temporarily permeable such that foreign DNA (e.g., plasmids) can be taken up. Electroporation can be used to introduce circular pieces of dsDNA into cells that are not naturally competent, such as gram-negative bacteria and some species of *Archaea*. This process yields artificially constructed transformants and is a key step in genetic manipulation.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 11.6

3) Compare and contrast generalized transduction with specialized transduction.
Answer: Transduction as a whole is the transfer of genetic information from one cell to another mediated by a virus. Generalized transduction allows for any portion of a cell′s genome to be transferred, but this process occurs at a very low frequency (roughly 10^{-6} to 10^{-8}). Specialized transduction transfers only genetic material adjacent to the integrated phage (prophage), however its transfer efficiency is much higher than in generalized transduction.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.7

4) Consider the map shown below of a portion of a prokaryotic genome. How might the elements indicated affect the stability and structure of the genome? Assume that IS2 encodes for a transposase that catalyzes conservative transposition. Re-draw the genome map to illustrate your prediction if necessary.

| IS2 | *gene1 gene2 gene3* | IS2 | *geneA geneB geneC* | IS2 | *geneD* | IS2 | *geneX*

Answer: Answers should indicate that having multiple IS2 elements in the genome could result in the deletion and/or rearrangement of gene order by transposition events. Composite transposons can form if the genes between to insertion sequences move, or transpose, with the insertion sequences. For instance, *geneD* could be removed from this part of the genome and inserted elsewhere. OR *geneABCD* could all move together to another part of the genome through transposition. Similarly *gene123* could move together, or even *gene123* and *geneABCD* (but not *geneX*). If conservative transposition is the only possibility then the genes will move from one spot to another in the genome, but if replicative transposition occurred the genes could be copied to another part of the genome.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.11

5) Discuss the importance of homologous recombination and transposition in natural horizontal gene transfer and evolution. Be sure to define all of the terms you use and connect them together logically in your answer.
Answer: Natural horizontal gene transfer is the movement of genes between cells that are not direct descendants of each other. Three mechanisms (transduction, transformation, and conjugation) can transfer genetic material between cells. Once the genetic material is in a cell, it is not necessarily passed down to daughter cells unless it can replicate on its own or is integrated into the genome. Homologous recombination combines similar DNA molecules from two different sources, thus homologous recombination aids in the integration of transferred genetic material into the recipient cell. Transformation also can move segments of DNA between two genetic elements, thus transformation also aids in the movement of genes from temporary or separate genetic elements into the genome of a recipient. Without homologous recombination and transposition, horizontal gene transfer would result in fewer permanent or heritable genetic changes. Thus without homologous recombination and transposition, horizontal gene transfer would have a lesser effect on evolution.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.11

6) Why is the outcome of a base-pair substitution mutation variable? List and describe the possible outcomes of a base-pair substitution mutation in your answer.
Answer: The outcome of a base-pair substitution mutation is variable because multiple codons can code for the same amino acid. For example, ACC, ACG, and ACU all encode for alanine (codon examples do not necessarily have to be correct, but should convey the concept). Often the third position of the codon can change without changing the amino acid. In addition, some amino acids have similar properties (e.g., leucine and isoleucine), thus changing particular amino acids may not have much of a phenotypic affect. There are three possible outcomes of a point mutation. Silent mutations can have slightly modified protein structures or unmodified ones, as long as the nucleotide change does not result in any change of the phenotype. Missense mutations are more restricted in that a nucleotide change results in a particular amino acid (codon) change. Although a missense mutation will change the amino acid sequence, this again may or may not change the structure of the protein and the phenotype. A nonsense mutation is the most specific of all where the nucleotide sequence now contains an inserted non-native stop site. The premature halt of protein synthesis will result in a shortened protein structure and will likely change the phenotype of the organism.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.2

7) How does the SOS system simultaneously fix damage DNA and increase the mutation rate? How does the SOS system increase the survival of microorganisms?
Answer: The SOS system causes the expression of a number of proteins, and specifically DNA polymerases, that fix lesions in DNA molecules but are prone to errors or do not use templates at all to synthesize new DNA. This allows the organism to survive DNA damage by fixing large lesions or breaks in the DNA but also results in mutations. Even under some conditions that do not result in a lot of DNA damage, the SOS system proteins can increase mutation rate. A slight increase in mutation rate may increase the probability of an advantageous mutation in one cell (even if many other cells are disadvantaged or killed by mutations). The cell with the advantageous mutation would be better able to survive the stress and could go on to replicate and take over the population, thus increasing the survival of the microorganism.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.4

8) A mutant you made in the laboratory with 5-bromouracil suddenly regains the wild-type phenotype. You discuss this phenomenon with your advisor and colleagues, and they suggest you try transposon mutagenesis to avoid this problem and create stable mutants. Why would mutants created with 5-bromouracil be more likely to regain the wild-type phenotype than mutants created via transposon mutagenesis?
Answer: Because a point mutation involves only a single base pair change, reversion is quite common relative to fixing an insertion of several nucleotides. Also common in *Bacteria* are suppressor mutations, which suppress the initial mutation and restore the wild-type phenotype (but not genotype) of a point mutation. However no analogous mechanism works to fix large-scale insertions or deletions.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.11

9) The CRISPR system is said to protect or preserve genome integrity in prokaryotic cells by acting as a type of ″immune system.″ Explain how the CRISPR system works and its limitations.
Answer: The CRISPR (clustered regularly interspaced short palindromic repeats) system is based on a region of the genome that contains DNA sequences from previously encountered foreign DNA separated by short palindromic repeats. The CRISPR region is translated into a long RNA molecule and then cleaved at each palindromic repeat to create short RNA molecules (crRNAs) that are complementary to foreign nucleic acids. If the crRNA binds to a piece of nucleic acid, associated proteins called Cas proteins will destroy the foreign nucleic acid and prevent it from recombining with the genome or replicating. The CRISPR system only works if the foreign DNA has homology with the DNA sequences in the CRISPR region of the genome. This means that novel phages or other DNA sequences would not be recognized or destroyed. Another limitation is that the CRISPR system might destroy DNA that contains useful genes or sequences if it were allowed to recombine with the genome. This could slow down evolution and adaptation.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 11.12

10) How might mutations in the *dnaQ* sequence, whose enzyme is involved in DNA proofreading, be beneficial for a microbe in a highly competitive and dynamic community?
Answer: Mutations in *dnaQ* would result in an increase in mutation rate; however unrepaired mutations are not always lethal. In this situation, mutagenesis might be thought of as creating diversity of a microbe′s genome. Genetic diversity might create a genotype different enough that a new phenotype arises. If for example the new phenotype is an ability to catabolize a particular compound, this could provide the cell with the ability to survive using a different substrate for energy when others are limited. Also, if a certain cell surface component is the target for phage infection and lysis, a different gene sequence could modify the surface component to perhaps enhance survivability in the environment.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 11.4